

Table 7.1 Market Impact Parameters by Trade Strategy Definition

	a_1	a_2	a_3	a_4	b_1	non- R^2
<i>All Data</i>						
POV	708	0.55	0.71	0.50	0.98	0.41
Trade Rate	534	0.57	0.71	0.35	0.96	0.41
Trade Schedule	656	0.48	0.45	1	0.90	0.38
<i>Large Cap Sample</i>						
POV	687	0.70	0.72	0.35	0.98	0.42
Trade Rate	567	0.72	0.73	0.25	0.96	0.42
Trade Schedule	707	0.59	0.46	1	0.90	0.37
<i>Small Cap Sample</i>						
POV	702	0.47	0.69	0.60	0.97	0.42
Trade Rate	499	0.49	0.69	0.40	0.97	0.42
Trade Schedule	665	0.42	0.47	1	0.90	0.39

Equation 7.16) and for each of the data samples: all data, large cap, and small cap categories. These results are shown in Table 7.1.

As expected, the non-linear R^2 statistics are almost equivalent for the POV and trade rate strategies since there is a near one-to-one relationship between POV and α , especially for realistic percentages of volume levels (e.g., $POV < 40\%$). Additionally, the trade schedule non-linear R^2 is just slightly lower than POV and trade rate strategies, which implies the trade schedule formulation provides reasonable results.

Comparison of parameter values, however, from the different models is not the preferred process to evaluate models. As we showed in Chapter 5, models could have seemingly different parameter sets yet provide the same cost estimates. Then the easiest way to compare models is through cost estimates for various sizes and strategies.

Our analysis consisted of comparing costs for sizes from 1% ADV to 35% ADV for a full day VWAP strategy and an equivalent $POV = 20\%$ strategy. We used the parameters for the full universe category and a volatility = 30% for the comparison test. The results are shown in Figure 7.2 and show that these results are consistent under the various model forms. Readers are encouraged to verify these calculations and to compare the models using the parameters from large cap and small cap data sets for different strategies.

Figure 7.2a compares market impact estimates for a VWAP strategy using POV, trade rate, and the trade schedule. Notice that the POV rate